

KAVITHA JEWELLERY

DEVELOPER TECHNICAL HANDOVER & ARCHITECTURE SPECIFICATION

Version 1.0 | Kerala 22K Gold Retail & Onam Campaign Platform | Confidential Developer Brief

1. PROJECT SUMMARY & TECH STACK

Kavitha Jewellery is a responsive, high-performance web application designed for a luxury 22K Gold & Diamond retail brand in Kerala. It combines an e-commerce catalog with a Live Gold Rate Price Breakdown Engine, a gamified Onam Campaign Voucher System with BSNL DLT OTP Verification, an In-Store Staff Redemption POS Terminal, and a Protected Admin Portal.

Component	Technology & Library	Role / Purpose
Frontend Framework	React 19 + TypeScript (Strict)	Core reactive UI and state architecture
Build System	Vite 6 (@tailwindcss/vite)	Ultra-fast HMR and production bundle bundling
Styling & Design	Tailwind CSS v4 + Playfair / Inter	Kerala luxury brand aesthetic with custom color palette
Animation & Polish	Motion (motion/react)	Smooth wheel spins, modals, toasts, tab transitions
Icons	Google Material Symbols + Lucide	E-commerce, hallmark, security, and retail icons
Deployment Target	Cloud Run / Static Container (Port 3000)	Zero-config Node / SPA runtime container environment

2. APPLICATION VIEWS & USER JOURNEYS

View File	Primary Responsibilities & Functional Features
HomeView.tsx	Hero banners, Live Gold Rate Ticker, Featured Collections, Trust Badges (BIS 916, 100% Exchange).
CatalogView.tsx	Filterable catalog with Purity (22K/18K/14K), Weight Range, Category tabs, Price Range, and Instant Search.
PdpView.tsx	Dynamic Price Breakdown Engine: (Gold Weight × $\frac{1}{2}$ Current Rate) + Making Charges + Wastage + Hallmarking + 3% GST. Includes Customer 5-Star Reviews & WhatsApp concierge.
CartView.tsx	Live synchronized shopping cart, gift notes, and BSNL DLT OTP verified checkout modal (CheckoutOtpModal.tsx).
OnamCampaignView.tsx	Gamified Lucky Gold Wheel with fraud-resistant OTP verification and instant voucher issuance.
StaffRedemptionView.tsx	In-Store POS Terminal: Phone lookup, voucher status verification, making charge discount rules, and mark as REDEEMED.
AdminCampaignView.tsx	Protected Admin Management: Real-time Gold Rate editor, Quota Pool Manager, DLT SMS Config, VIP Voucher Generator, Margin Simulator.

3. DATA PERSISTENCE & STORAGE KEYS

All state currently operates out-of-the-box using structured browser persistence. Developers can replace these modular stores with PostgreSQL, Firebase Firestore, or Supabase:

Storage Key	Type	Description / Business Logic
kavitha_live_gold_rate	localStorage	Global 22K gold rate (INR/g). Changes propagate instantly across all product prices.
kavitha_shopping_cart	sessionStorage	Active cart items, quantities, and selected purities.
kavitha_onam_coupons	localStorage	Array of issued vouchers, customer phone mappings, issued dates, and redeemed statuses.

Storage Key	Type	Description / Business Logic
kavitha_coupon_pool_config	localStorage	Dynamic campaign quota pools (e.g. ₹50,000, ₹25,000 caps).
kavitha_bsnl_dlt_config	localStorage	BSNL DLT Entity ID, 6-letter Header (KAVITH), Approved Template ID, and SMS Gateway mode.
kavitha_reviews_prod_{id}	localStorage	Customer reviews, ratings (1-5★), verified buyer tags, and product SKU.

4. ADMIN CAPABILITIES & BSNL DLT TELECOM INTEGRATION

- 🔧 Master Gold Rate Control: Updating the 22K rate in the admin portal immediately recalculates catalog pricing and checkout totals.
- 🔧 Quota Allocation Engine: Prevents financial budget overrun by capping high-tier reward redemptions.
- 🔧 BSNL DLT Compliance: Configured for TRAI mandate with Principal Entity ID, Header (KAVITH), and 100% compliant SMS templates.
- 🔧 SMS Gateway Support: Supports Fast2SMS, MSG91, TextLocal, Twilio, and Simulated Sandbox modes with an integrated live test utility.
- 🔧 VIP Voucher Generator & Margin Simulator: Issue custom direct vouchers to dignitaries and calculate making charge discount margins.

5. CODEBASE MAP & RECOMMENDED NEXT STEPS

Directory / File	Contents & Action Items for Backend Developer
src/data/products.ts	Catalog & price calculations. Connect to database endpoints: GET /api/products, GET /api/products/:id.
src/data/campaignData.ts	Campaign rules & quota engine. Connect to POST /api/vouchers/issue, POST /api/vouchers/redeem.
src/data/dltSmsConfig.ts	DLT configuration & SMS triggers. Provide live Gateway API Key (Fast2SMS/MSG91) in Admin or ENV.
src/data/reviewsData.ts	Customer reviews store. Connect to GET /api/reviews?productId=... and POST /api/reviews.
src/views/CartView.tsx	Checkout flow. Ready for Razorpay / Cashfree / Stripe payment gateway SDK hookup.
src/types.ts	Master TypeScript definitions (Product, Coupon, Review, CartItem, Order, User).

QUICK START COMMANDS FOR DEVELOPERS:

```
npm install && npm run dev (Port 3000) && npm run lint && npm run build
```